INCH-POUND
MIL-DTL-24749C(SH)
21 May 2019
SUPERSEDING
MIL-DTL-24749B(SH)
w/INT. AMENDMENT 1
15 March 2016
MIL-DTL-24749B(SH)
28 January 2014

DETAIL SPECIFICATION

GROUNDING STRAPS AND STUDS, ELECTRICAL BONDING

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for MIL-STD-1310 grounding straps and welding studs for electrical bonding, hereafter identified as bond straps and bosses, used in electromagnetic compatibility (EMC), electromagnetic pulse (EMP) mitigation, and electrical safety.

1.2 Classification. The bond straps and bosses contained in this specification are classified into the following types, meeting general characteristics as follows (see 6.1 and 6.2).

1.2.1 Bond strap types.

a. Type I – Corrosion-resistant steel (CRES) 316 wire rope with mounting lugs
b. Type II – Flat CRES 316 strip with mounting holes on each end
c. Type III – Flat copper strip with mounting holes on each end
d. Type IV – Flat CRES 316L/nickel 200 braid with mounting lugs

1.2.2 Boss types.

a. AL – Aluminum/CRES 316L
b. AL-SGC – Aluminum/CRES 316L stud gun compatible
c. CR – CRES 316L
d. CR-SGC – CRES 316L stud gun compatible

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5975

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

1.2.3 Bond strap sizes. Unless otherwise specified (see 6.2), bond strap sizes are as specified in table I, table II, and table III.

TABLE I. Type I bond strap standard sizes.

Standard size code	Length	Hole diameter
A	6.0	13/32 (0.406)
B	12.0	13/32 (0.406)
C	18.0	13/32 (0.406)

NOTES:
1. Dimensions are in inches.
2. Length is end-to-end.
3. Lengths are ±0.250 inch.
4. Hole diameter tolerance is ±0.010 inch.

TABLE II. Types II and III bond strap standard sizes.

Standard size code	Length	Width	Hole diameter
A	3.0	1.0	13/32 (0.406)
B	6.0	1.5	13/32 (0.406)
C	9.0	2.0	13/32 (0.406)
D	12.0	2.5	13/32 (0.406)

NOTES:
1. Dimensions are in inches.
2. Length is ±0.250 inch hole-to-hole.
3. Width is ±0.125 inch.
4. Hole diameter tolerance is ±0.010 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

TABLE III. Type IV bond strap standard sizes.

Standard size code	Length	Hole diameter
A	6.0	13/32 (0.406)
B	12.0	13/32 (0.406)
C	18.0	13/32 (0.406)
D	6.0	9/32 (0.281)
E	12.0	9/32 (0.281)
F	18.0	9/32 (0.281)

NOTES:
1. Dimensions are in inches.
2. Nominal length is hole-to-hole ±5 percent.
3. Hole diameter tolerance is ±0.010 inch.

1.3 Part or identifying number (PIN). PINs to be used for bond straps and bosses acquired to this specification are created as follows (see 6.2):

M	**24749**	-	**X**	-	**X**	-	**X**	-	**X**	-	**X**	-	**X**
Prefix for Military Specification	Specification Number		Bond strap type (see 1.2.1)		Standard size code (see 1.2.3; use size code "N" if size is nonstandard)		Length (only for nonstandard sizes)		Width (only for types II and III bond straps in nonstandard sizes)		Hole diameter (only for types II and III bond straps in nonstandard sizes)		Boss type (only for type I bond straps, see 1.2.2)

EXAMPLE: M24749-II-B (type II bond strap, standard size B).

EXAMPLE: M24749-III-C (type III bond strap, standard size C).

EXAMPLE: M24749-I-A-AL-SGC (type I bond strap, standard size A; boss type aluminum/CRES 316L; stud gun compatible).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59569 - Braid, Wire (Copper, Tin-Coated, Silver-Coated, or Nickel Coated, Tubular or Flat)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

MIL-S-24149/3 - Studs, Welding, & Arc Shields (Ferrules); Type V, Class 1, 4, 5, 5A, Corrosion-Resistant Steel, for Direct Energy Arc Welding

MIL-J-24445 - Joint, Bimetallic Bonded, Aluminum to Steel

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

48 CFR 252.225-7009 - Restriction on Acquisition of Certain Articles Containing Specialty Metals

(Copies of this document are available online at www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A269/A269M - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service

ASTM A312/A312M - Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes

ASTM A342/A342M - Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM A492 - Standard Specification for Stainless Steel Rope Wire

ASTM A666 - Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar

ASTM B152/B152M - Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar

ASTM B928/B928M - Standard Specification for High Magnesium Aluminum-Alloy Products for Marine Service and Similar Environments

ASTM F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), samples shall be subjected to first article inspection in accordance with 4.4. When first article inspection has been previously accomplished and no changes in raw materials, manufacturing processing or procedures, or type of equipment have occurred, conformance inspection alone in accordance with 4.5 shall be conducted. Conformance inspection alone shall not be used to satisfy first article inspection of product for delivery without a previous first article inspection from the same type, size, material, and style produced at the same facility using the same purchased components, production processes, and equipment. First article inspection shall always be accomplished when there are changes to raw materials, manufacturing processing or procedures, or type of equipment.

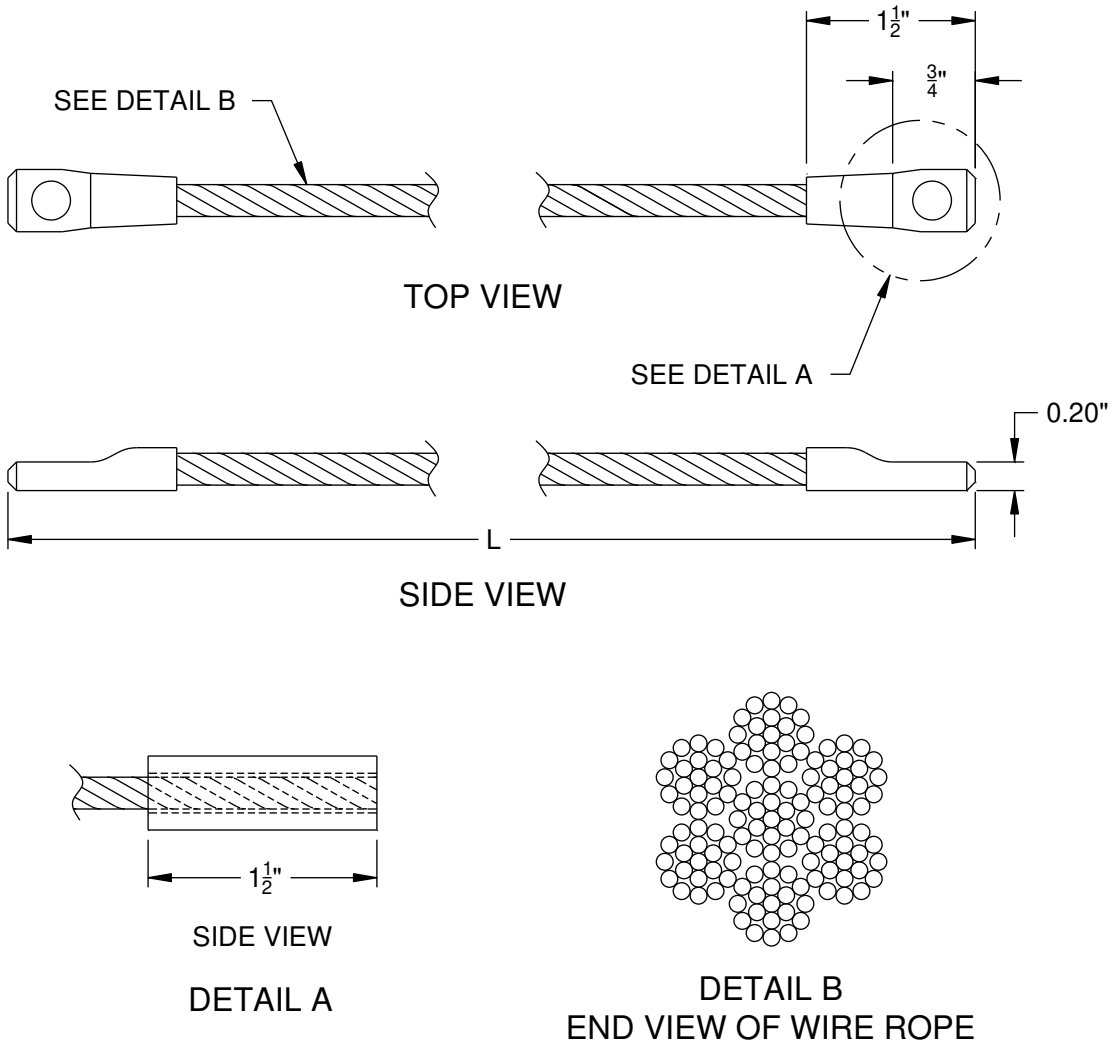
3.2 Materials, components, and construction. All materials, components, and construction methods shall conform to applicable specifications, standards, and figures specified herein. Items comprised of specialty materials (i.e., CRES 316/316L) shall be in accordance with CFR 252.225-7009.

3.2.1 Type I bond strap. A type I bond strap shall consist of the following three components:

a. One annealed CRES 316, UNS S31600, ¼-inch (0.250-inch) diameter stranded wire cable or wire rope.

b. Two end lugs formed from CRES 316L, UNS S31600, schedule 40S, pipe (see figure 1, detail A).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.



NOTES:

1. Insert stranded wire through entire 1.500-inch length of CRES 316L pipe as shown in detail A before compression.
2. Compress the end ¾ (0.750) inch of the pipe to 0.200±0.025-inch thickness. Ensure both mounting surfaces of the lug are smooth and flat and burnish the tip.
3. Drill a $^{13}/_{32}$ inch (0.406 inch) ±0.010-inch hole in each lug to facilitate ⅜-inch bolts.
4. Unless otherwise noted herein, all figure dimensions are in inches ±$^{1}/_{16}$ (0.0625) inch.
5. Bond strap length (L) is end-to-end.

FIGURE 1. Type I bond strap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

3.2.1.1 Stranded wire rope. Stranded wire rope shall be ¼-inch (0.250-inch) diameter CRES 316, UNS S31600, of 7-by-19 construction (see figure 1, detail B). Wire used to construct the wire rope shall be in accordance with ASTM A492. Wire rope of this construction shall consist of 6 outer strands of 19 wires each laid around a strand core of 19 wires. The 6 outer strands shall each consist of a layer of 6 wires laid around a center wire in a left-hand direction and a layer of 12 wires laid over the 7-wire strand in a left-hand direction. The strand core shall consist of a layer of 6 wires laid around a center wire in a right-hand direction and a layer of 12 wires laid around the 7-wire strand in the right-hand direction. The 6 outer strands shall be laid around the core in a right-hand direction. The length of lay of the inside layer of 6 wires in each of the 6 outer strands and the 1 strand core shall not exceed 60 percent of the outside layer of 12 wires in each strand. The length of lay of the outside layer of 12 wires in each of the 6 outside strands and the strand core shall not exceed 50 percent of the lay of the finished wire rope. The length of the lay of the finished wire rope shall be not greater than 2 inches nor less than 1.5 inches. The finished wire rope shall be annealed by the strand annealing process specified in 4.7.5 so that it has no residual martensitic or magnetic properties. The breaking strength of the fully annealed wire rope shall be between 2,000 and 3,000 pounds, and the minimum elongation shall be 30 percent in a 12-inch gauge length.

3.2.1.2 Lugs. Lugs shall be made of 0.405-inch outside diameter (OD) and 0.269-inch inside diameter (ID) CRES 316L, UNS S31603, schedule 40S, pipe in accordance with ASTM A312/A312M. The lugs shall be formed from the blank pipe as follows:

a. Prepare the lug by cutting off a section of pipe to 1½-inch length.

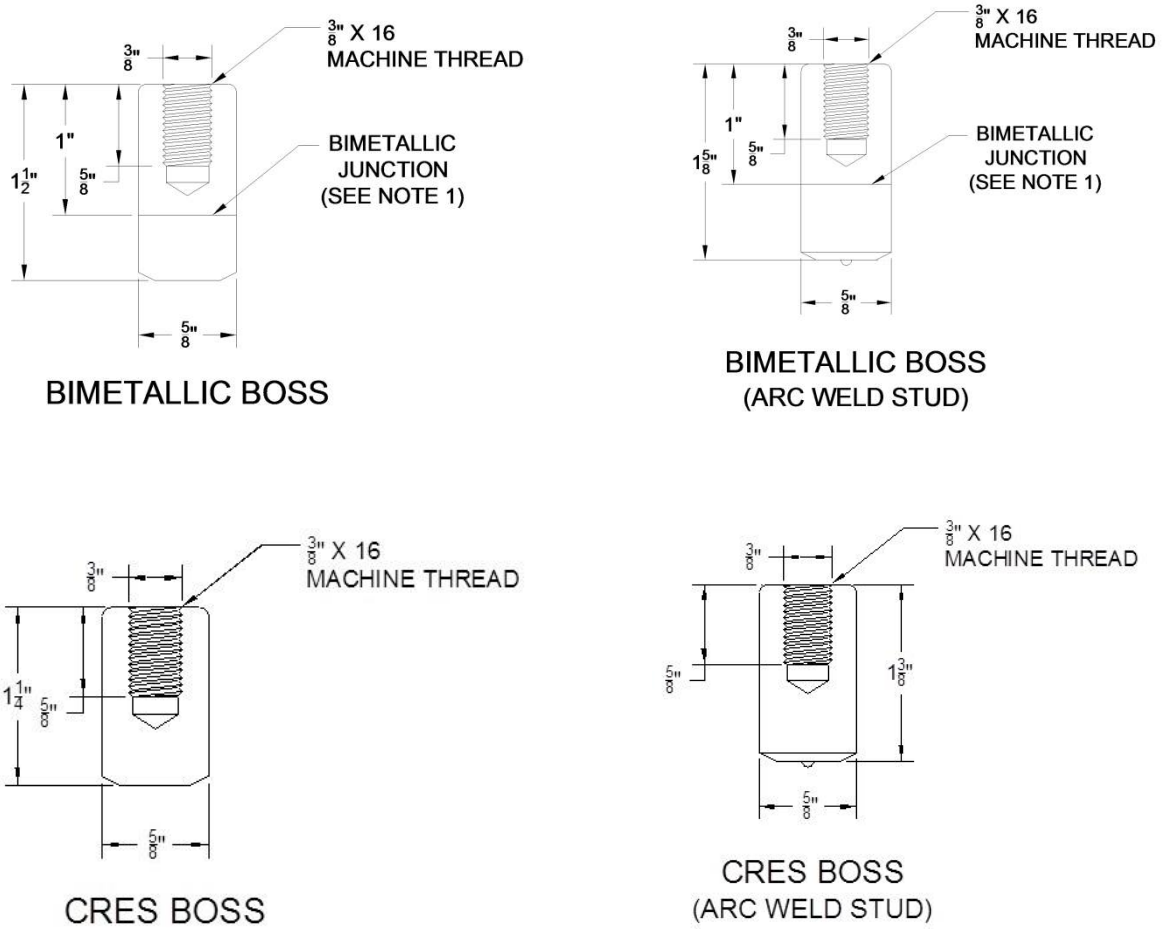
b. Insert stranded wire cable or wire rope through the entire 1½-inch length of the pipe.

c. Compress the last ¾ (0.750) inch of the pipe to 0.200±0.010-inch thickness and ensure that both mounting surfaces of the lug are smooth and flat.

d. Burnish the end to remove sharp edges and drill a $^{13}/_{32}$ (0.406)±0.010-inch hole in the flattened lug section to facilitate a ⅜-inch bolt.

3.2.2 Bosses. Bosses used with type I bond straps shall be ⅝ (0.625) inch in diameter, and shall either be CRES 316L, UNS S31603, in accordance with SAE-AMS-QQ-S-763, or bimetallic (CRES 316L/aluminum) in accordance with MIL-J-24445. Dimensions associated with bosses shall be as shown on figure 2. Bosses shall be either conventional or arc weld studs. Arc weld studs shall be in accordance with MIL-S-24149. Each boss shall contain a CRES 316 bolt (hex cap screw) as specified in 3.2.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.



NOTES:

1. Bolt holes shall not penetrate through the bond lines of the bimetallic transition layer.
2. All dimensions, except those for machine threads, are in inches $\pm 1/32$ (0.031) inch.
3. The manufacturer shall be responsible for designing boss weld ends that will provide a sound weld when used with the proper ferrule for arc-welding (see MIL-S-24149/3) or fillet welding.

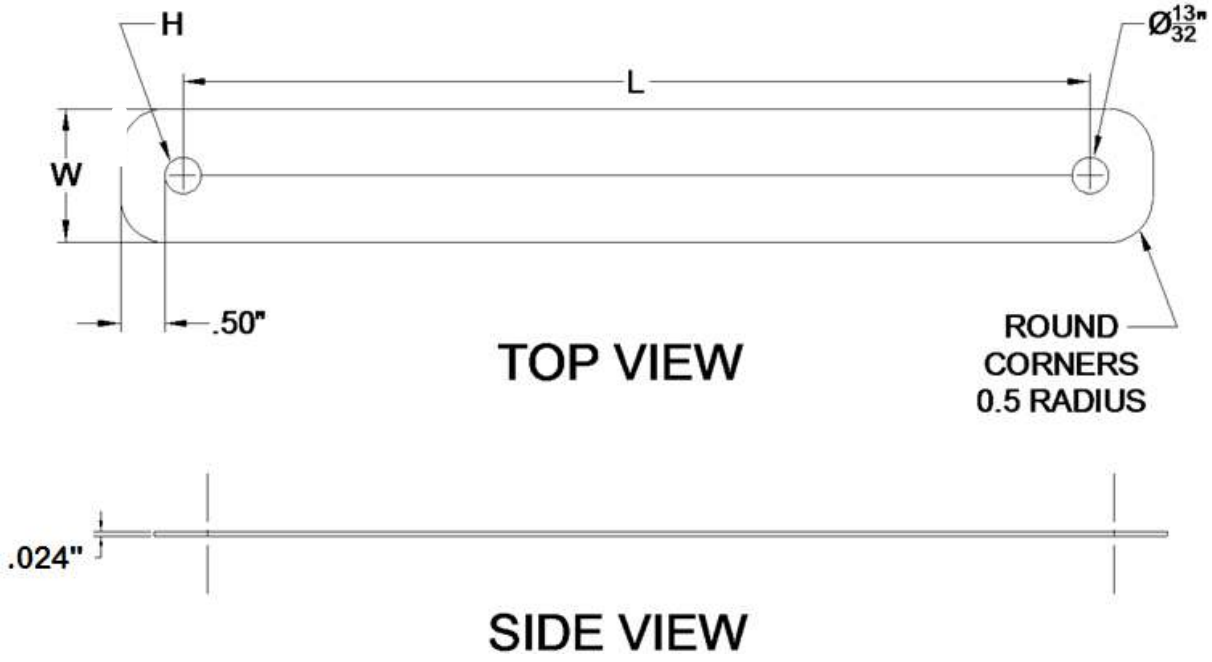
FIGURE 2. Bosses.

3.2.2.1 Bimetallic boss. The foot of the aluminum boss shall be manufactured from aluminum alloy 5083, 5086, or 5456, in accordance with ASTM B928/B928M. The foot shall be bonded to CRES 316, UNS S31600, to form the CRES 316/aluminum boss. An interstitial layer of an appropriate metal may be used to enhance strength and weldability. Chromium, tantalum, molybdenum, or titanium shall be used if an interstitial layer is used.

3.2.2.2 Boss bolts. Bolts used in conjunction with bosses shall be fabricated from CRES 316, UNS S31600. The bolt for attachment of the type I bond strap to the boss shall be ⅜-16UNS by ¾ (0.750) inch in accordance with ASTM F593 or MIL-DTL-1222.

3.2.3 Type II bond strap. Type II bond straps shall be manufactured of annealed CRES 316, UNS S31600, in accordance with ASTM A240/A240M or ASTM A666. The thickness of strap shall be 0.024±0.003 inch, and it shall be fabricated as shown on figure 3. The length-to-width ratio of type II bond straps shall not exceed 5 to 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.



NOTES:

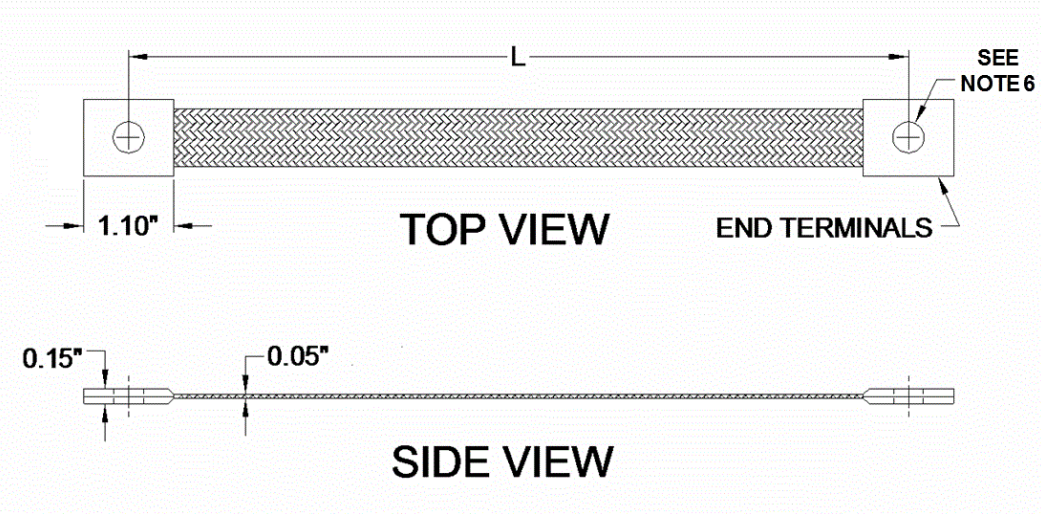
1. Length (L), width (W), and hole diameter (H) may be specified in non-standard sizes (see 1.3 and 6.2).
2. See table II for type II and type III bond strap standard sizes.
3. Unless otherwise noted in 3.2.3 or 3.2.4, all dimensions are in inches ±0.010 inch.

FIGURE 3. Types II and III bond strap fabrication details.

3.2.4 Type III bond strap. Type III bond straps shall be manufactured of UNS C11000 (electrolytic tough pitch), temper H00 (cold-rolled), flat copper strip in accordance with ASTM B152/B152M. The thickness of strap shall be 0.024±0.003 inch, and it shall be fabricated as shown on figure 3. The length-to-width ratio of type III bond straps shall not exceed 5 to 1.

3.2.5 Type IV bond strap. Type IV bond straps shall be fabricated as shown on figure 4 and shall consist of one wire braid strap and two end terminals. Ensure that both mounting surfaces of the terminals are smooth and flat and burnish the end to remove sharp edges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.



NOTES:

1. End terminals shall be formed from CRES 316L welded-wall tube of 0.625-inch OD and a wall thickness of 0.065 inch ±10 percent, in accordance with ASTM A269/A269M, cut to 1.100±0.020-inch length.
2. The braid shall be flattened to 0.05±0.010 inch, and the braid ends shall be placed completely inside the pipe ends.
3. The pipe shall be compressed to 0.150±0.010 inch to hold fast the braid.
4. Length (L) may be specified in nonstandard sizes (see 1.3 and 6.2).
5. Drill or punch a $^{13}/_{32}$ (0.406)±0.010-inch hole in each lug after end terminal installation to facilitate ⅜-inch bolts (for size codes A through C), or a $^{9}/_{32}$ (0.281)±0.010-inch hole in each lug to facilitate ¼-inch bolts (for size codes D through F).
6. See table III for type IV bond strap standard sizes.

FIGURE 4. Type IV bond strap fabrication details.

3.2.5.1 Braid. The bond strap braid shall be 50 percent CRES 316L wires and 50 percent nickel 200 wires of 36 American Wire Gauge (AWG). The braid shall be constructed in accordance with A-A-59569, PIN 36*1000, using 6 strands of 36 AWG nickel 200 wire and 6 strands of 36 AWG CRES 316L stainless steel, woven on a 64 carrier machine with 768 ends to produce an approximately 1-inch wide flat braid strap.

3.2.5.2 End terminal. End terminals shall be annealed, welded-wall, or seamless CRES 316L tube, in accordance with ASTM A269/A269M. Dimensions shall be 1.100±0.020 inch in length with an OD of 0.625 inch and a nominal wall thickness of 0.065 inch. The end of the braid shall lay 100 percent inside the tube and the tube shall be flattened to 0.150±0.010 inch onto the braid. Drill or punch a $^{13}/_{32}$ (0.406) ±0.010-inch hole in each lug to facilitate ⅜-inch bolts (for size code A through C), or a $^{9}/_{32}$ (0.281) ±0.010-inch hole in each lug to facilitate ¼-inch bolts (for size code D through F).

3.3 Performance requirements.

3.3.1 Bond joint strength and uniformity. The bimetallic boss joint shall be tested as specified in 4.7.1 and shall meet the requirements for both gas and electric arc weld studs in accordance with MIL-J-24445.

3.3.2 Pull test. When tested as specified in 4.7.2, types I and IV bond straps shall withstand a load of 300 pounds and be maintained within ±50 pounds for 3 minutes. Wire rope and braid shall not be pulled from lugs and no visible wire breakage shall occur.

3.3.3 Conformability and flexibility. Types I and IV bond straps shall be tested as specified in 4.7.3. Wire rope and braid shall not be pulled from lugs and shall exhibit no visual evidence of wire breakage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

3.3.4 Electrical resistance. Types I and IV bond straps shall be tested as specified in 4.7.4. The type I strap's direct current (DC) resistance shall be not greater than 1.15 milliohm per 1 inch of strap length. The type IV strap's DC resistance shall be not greater than 0.5 milliohm per 1-inch length.

3.3.5 Permeability. The wire rope in type I bond straps shall have less than 1.002 relative permeability when tested as specified in 4.7.5.

3.4 Identification. Type I bond straps shall be marked with the manufacturer's Commercial and Government Entity (CAGE) code. Types II, III, and IV bond straps shall be marked with an identifying legend consisting of the specification number with revision letter, the PIN (see 1.3), and the manufacturer's CAGE code (see 6.2). The marking shall be made permanent by use of metal stamp, indelible ink, or electro-etch, with a character height of not less than 0.08 inch.

3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.6 Workmanship. MIL-HDBK-454 should be used as a guideline to ensure the use of best practices for manufacturing bond straps. The materials used in the manufacture of bond straps and bosses shall be free from all defects that could adversely affect installation, appearance, or long-term use. The mounting hole of flat plate straps and lugs of wire rope strap and flat braid straps shall be free from blistering, burrs, cracks, laps, peeling, pitting, sharp edges, slag, and other foreign material that could affect welding or bond strap installation (see 4.6).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

4.2 Inspection conditions. All inspections shall be performed in accordance with the test conditions specified in 4.4 and 4.5.

4.3 Lot. An inspection lot is defined as items of the same type, size, material, and style produced at the same facility using the same purchased components, production processes, and equipment.

4.4 First article inspection. A first article inspection shall be performed on each type of bond strap or boss when a first article inspection is required (see 3.1). First article inspection shall include the tests specified in table IV in accordance with the sampling specified in table V and the tests of 4.6 through 4.7.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

TABLE IV. First article inspection.

Inspection/test	Requirement	Test
Type I bond strap		
Visual inspection	3.2.1 3.4 3.6	4.6
Pull test	3.3.2	4.7.2
Conformability and flexibility	3.3.3	4.7.3
Electrical resistance	3.3.4	4.7.4
Permeability	3.3.5	4.7.5
Type I bond strap bosses	3.3.1	4.7.1
Types II and III bond straps		
Visual inspection	3.2.3 3.2.4 3.4 3.6	4.6
Type IV bond strap		
Visual inspection	3.2.5 3.4 3.6	4.6
Pull test	3.3.2	4.7.2
Conformability and flexibility	3.3.3	4.7.3
Electrical resistance	3.3.4	4.7.4
Packaging (all types)	5.1	--

TABLE V. First article inspection sampling.

Lot size	Sample size	
	Major defect characteristics	Minor defect characteristics
20 to 25	15	3
26 to 50	20	5
51 to 90	20	6
91 to 150	20	7
151 to 280	20	10
281 to 500	47	11
501 to 1,200	47	15
1,201 to 3,200	53	18
3,201 to 10,000	68	22
10,001 to 35,000	77	29
35,001 to 150,000	96	29

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

4.4.1 Sample size. A minimum of 20 sample units from each standard size shall be submitted to be eligible for conformance inspection testing.

4.4.2 First article inspection sampling. As a minimum, a sample quantity of completed bond straps shall be randomly selected for first article inspection in accordance with the sampling plan specified in table VI as determined by the defect characteristic indicated in table IV. Sample size depends on the lot size as shown in table V. If one or more defects are found in any sample, the entire lot shall be rejected.

TABLE VI. Classification of defects.

Category	Characteristic
Major	
101	The bond straps do not meet the material and construction requirements (see 3.2).
102	Type I bond strap bosses do not meet bond joint strength and uniformity requirements (see 3.3.1).
103	Type I bond straps do not meet pull test requirements (see 3.3.2).
104	Type I bond straps do not meet conformability and flexibility requirements (see 3.3.3).
105	Types I and IV bond straps do not meet electrical resistance requirements (see 3.3.4).
106	Type I bond straps do not meet permeability requirements (see 3.3.5).
107	The bond straps do not meet workmanship requirements (see 3.6).
Minor	
201	Type IV bond straps do not meet pull test requirements (see 3.3.2).
202	Type IV bond straps do not meet conformability and flexibility requirements (see 3.3.3).
203	The bond straps do not meet identification requirements (see 3.4).
204	The packaging requirements are not met (see 5.1).
NOTE: 1. See 6.9.	

4.5 Conformance inspection. Conformance inspection shall be performed once a successful first article inspection has been accomplished (see 3.1, 4.4.1, and 6.4). Conformance inspection shall include the tests specified in table VII in accordance with the sampling specified in table VIII and the tests of 4.6 through 4.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

TABLE VII. Conformance inspection.

Inspection/test	Requirement	Test
Type I bond strap		
Visual inspection	3.2.1 3.4 3.6	4.6
Electrical resistance	3.3.4	4.7.4
Type I bond strap bosses	3.3.1	4.7.1
Types II and III bond strap	3.3.1	4.7.2
Visual inspection	3.4 3.6	4.6
Type IV bond strap		
Visual inspection	3.2.5 3.4 3.6	4.6
Electrical resistance	3.3.4	4.7.4
Packaging (all types)	5.1	--

TABLE VIII. Conformance inspection sampling.

Lot size	Sample size	
	Major defect characteristics	Minor defect characteristics
20 to 25	15	3
26 to 50	20	5
51 to 90	20	6
91 to 150	20	7
151 to 280	20	10
281 to 500	47	11
501 to 1,200	47	15
1,201 to 3,200	53	18
3,201 to 10,000	68	22
10,001 to 35,000	77	29
35,001 to 150,000	96	29

4.5.1 Conformance inspection sampling. As a minimum, a sample quantity of completed bond straps shall be randomly selected for conformance inspection in accordance with the sampling plan specified in table VII as determined by the defect characteristic indicated in table VI. Sample size depends on the lot size, as shown in table VII. If one or more defects are found in any sample, the entire lot shall be rejected.

4.6 Visual inspection. A visual inspection shall be performed on all samples prior to testing. The inspection shall determine that all samples are acceptable products for testing and that obvious defects are identified prior to the start of tests. The visual inspection shall determine conformance to, but shall not be limited to, the requirements specified in 3.2, 3.4, and 3.6. Sample products having defects that may affect the result of testing shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

4.7 Methods of test.

4.7.1 Bimetallic joint. Bimetallic joint testing shall be performed to meet the requirements of 3.3.1 as follows:

a. Four specimens of each bimetallic boss production run of up to and including 400 items shall be randomly selected for testing.

b. Two of the specimens shall be sectioned, polished, etched, and examined at 100× magnification for metallographic analysis. The weld heat-affected zone (HAZ) shall be uniform across the bond line. The bond shall be uniform and free of any indications of unbonded areas and shall be in accordance with the inclusion acceptance standards of ASTM B928/B928M for aluminum or SAE-AMS-QQ-S-763 for CRES 316L.

c. The remaining specimens shall be subjected to the bend test and the as-clad ram tensile test in accordance with MIL-J-24445.

d. If any specimen fails, the entire lot shall be rejected.

4.7.2 Pull test. Types I and IV bond straps shall be placed in a tensile testing configuration with the load distribution from lug to lug. The load shall be 300 pounds and maintained within ±50 pounds for 3 minutes. The strap shall meet the requirements specified in 3.3.2.

4.7.3 Conformability and flexibility. Types I and type IV straps shall be placed over a 1-inch diameter pin and shall be bent 180 degrees back and forth over the pin for 50 cycles. The bending force shall be no greater than 5 pounds. The strap shall meet the requirements specified in 3.3.3.

4.7.4 Electrical resistance. Types I and IV bond straps shall be tested for DC resistance from lug hole to lug hole end. The strap's DC resistance shall meet the requirements specified in 3.3.4.

4.7.5 Permeability. Type I bond straps shall be annealed and tested in accordance with ASTM A342/342M. There shall be a hydrogen-protective atmosphere within the annealing tube to eliminate the chance of oxidization of the surface of the wire rope and premature failure of the grounding strap due to corrosion. The annealing temperature shall be 2,000±50 °F. Throughput speed shall be 2.5±0.5 feet per minute. The straps' permeability shall be not greater than that specified in 3.3.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bond straps and bosses covered by this specification are intended for, but not limited to, shipboard bonding and grounding applications and as specified for Secure Electrical Information Processing (SEIP) equipment bonding and grounding. Recommended usage should be as follows:

a. Type I – for bonding items in weather-exposed areas where corrosion is a concern.

b. Type II – for lower-impedance bonding items in weather-exposed areas where corrosion is a concern.

c. Type III – for bonding items in non-weather-exposed areas.

d. Type IV – for bonding items in non-weather-exposed areas where corrosion is a concern. Type IV is also for use below deck across sound-dampening devices, in grounding or bonding of ¼-inch stud-mounted equipment, or in grounding or bonding of portable electrical equipment.

e. AL and AL-SGC bosses – for attaching type I straps to aluminum structures.

f. CR and CR-SGC bosses – for attaching type I straps to steel structures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and quantity of bond straps and bosses required (see 1.2 and 6.6).

c. Whether nonstandard bond strap dimensions in accordance with MIL-STD-1310, type V, are required (see 1.2.3).

d. Bond strap or boss PIN, when applicable (see 1.3).

e. When first article inspection is required (see 3.1).

f. Whether identification markings should include the manufacturer's CAGE code, the item's PIN, or both (see 3.4).

g. Packaging requirements (see 5.1).

h. Quantity of each bolt.

i. Any special instructions or remarks.

6.3 First article. When inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should include first article sample, a first production item, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and the disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Conformance inspection on existing units. Units that have successfully undergone first article inspection (see 4.4), but were not purchased by the Government at that time, do not need to undergo first article inspection again if the Government purchases them at a later time (see 3.1). Units fitting this description should undergo conformance inspection to ensure that they still meet minimal first article requirements herein (see 4.5).

6.5 Sub-contracted material and parts. The packaging requirements of the reference documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into bond straps and bond strap assemblies and lose their separate identity when bond straps and bond strap assemblies are shipped.

6.6 Preservation. Bonding straps of the same type and length, or bosses of the same foot material, in the quantity specified (see 6.2) should be unit-protected in accordance with ASTM D3951.

6.7 Packing. Commercial packing should conform to ASTM D3951.

6.8 Strap information. Information should be furnished with the bond straps and should include the manufacturer's lot number, the part number, the length, and the marking information.

6.9 Definitions.

6.9.1 Major defect. A defect likely to result in failure or reduce materially the usability of the product for its intended purpose.

6.9.2 Minor defect. A defect not likely to reduce materially the usability of the product for its intended purpose, or a departure from established standards having little bearing on the effective use or operation of the product.

6.10 Subject term (key word) listing.

Bond strap

Boss

Electromagnetic

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.

Electromagnetic compatibility

Electromagnetic pulse mitigation

Nut, sleeve

Stud, welding

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Reviewer:
DLA – GS

Preparing Activity:
Navy − SH
(Project 5975-2019-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:41Z
Check the source to verify that this is the current version before use.